

Lecture 4: Confidence intervals

Economics 527: Econometric Methods

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Normal estimator

Normal regression model

- **Assumption 1.** $Y = X\beta + U$ for some $\beta \in \mathbb{R}^k$.
- **Assumption 2.** $E[U | X] = 0$.
- **Assumption 3.** For some $\sigma^2 > 0$:

$$\begin{aligned}\text{Var}(U | X) &= E[UU^\top | X] \\ &= \sigma^2 I_n\end{aligned}$$

- **Assumption 4.** $\text{rank}(X) = k$.
- **Assumption 5.**

$$U | X \sim N\left(\underbrace{0}_{E[U|X]}, \underbrace{\sigma^2 I_n}_{\text{Var}(U|X)}\right)$$

Distribution of $\hat{\beta}$

- We will use:

$$\begin{aligned}V &\sim N\left(\underbrace{\mu}_{p \times 1}, \underbrace{\Sigma}_{p \times p}\right) \\ \Rightarrow \underbrace{A}_{m \times p} V + \underbrace{b}_{m \times 1} &\sim N(A\mu + b, A\Sigma A^\top)\end{aligned}$$

- Write $\hat{\beta}$ as a linear function of U and apply the result:

$$\begin{aligned}\hat{\beta} &= (X^\top X)^{-1} X^\top Y \\ &= \beta + \underbrace{(X^\top X)^{-1} X^\top}_A U \\ \Rightarrow \boxed{\hat{\beta} | X &\sim N(\beta, \sigma^2 (X^\top X)^{-1})}\end{aligned}$$

One coefficient $\hat{\beta}_1$

- Apply the same result to the first coefficient:

$$\begin{aligned}\hat{\beta}_1 | X &= \underbrace{[1 \ 0 \ \dots \ 0]}_{e_1^\top} \hat{\beta} | X \\ &\sim N\left(e_1^\top \beta, \underbrace{e_1^\top}_{1 \times k} \underbrace{\overbrace{\sigma^2 (X^\top X)^{-1}}^{\text{Var}(\hat{\beta}|X)}}_{k \times k} \underbrace{e_1}_{k \times 1}\right) \\ &= N(\beta_1, \text{Var}(\hat{\beta}_1 | X))\end{aligned}$$

Partitioned regression

- Regression on $X = \begin{bmatrix} X_1 & X_2 \end{bmatrix}$:

$$Y = \underset{1 \times 1}{\beta_1} \underset{n \times 1}{X_1} + \underset{n \times k_2}{X_2} \underset{k_2 \times 1}{\beta_2} + U$$

- Third lecture, the OLS coefficient on X_1 , with $M_2 = I_n - X_2(X_2^\top X_2)^{-1}X_2^\top$:

$$\hat{\beta}_1 = \frac{X_1^\top M_2 Y}{\underbrace{X_1^\top M_2 X_1}_{1 \times 1}}$$

Distribution of $\hat{\beta}_1$

- Under Assumptions 1 to 5:

$$\hat{\beta}_1 | X \sim N\left(\beta_1, \frac{\sigma^2}{X_1^\top M_2 X_1}\right)$$

Generic estimator $\hat{\theta}$

- The regression example in generic notation, conditional on X :

$$\theta = \beta_1, \quad \hat{\theta} = \hat{\beta}_1, \quad \omega^2 = \frac{\sigma^2}{X_1^\top M_2 X_1}$$

- Generic problem, an exactly normal estimator of θ :

$$\underset{1 \times 1}{\hat{\theta}} \sim N\left(\underset{1 \times 1}{\theta}, \underset{1 \times 1}{\omega^2}\right)$$

- ω^2 : known.
- In the regression example: σ^2 known $\implies \omega^2$ known.
- Probability that $\hat{\theta}$ equals θ :

$$P(\hat{\theta} = \theta) = 0$$

Confidence interval

Coverage

- Definition.** $CI_{1-\alpha}$, a confidence interval with coverage $1 - \alpha$, is a random interval such that:

$$P(\theta \in CI_{1-\alpha}) = 1 - \alpha$$

- Chosen in advance: $\alpha \in (0, 1)$, for example 0.01, 0.05, 0.1.
- Regression example:

$$P(\beta_1 \in CI_{1-\alpha} | X) = 1 - \alpha$$

- At least $1 - \alpha$ is acceptable, less is not:

$$\begin{aligned} P(\theta \in CI_{1-\alpha}) &\geq 1 - \alpha \\ P(\theta \in CI_{1-\alpha}) &< 1 - \alpha \quad \times \end{aligned}$$

Symmetric interval

- Symmetric around $\hat{\theta}$, with $c > 0$:

$$CI_{1-\alpha} = [\hat{\theta} - c, \hat{\theta} + c]$$

- Length of the interval:

$$\begin{aligned} \text{length} &= \hat{\theta} + c - (\hat{\theta} - c) \\ &= 2c \end{aligned}$$

- Problem: choose c for coverage $1 - \alpha$:

$$\begin{aligned} 1 - \alpha &= P(\theta \in CI_{1-\alpha}) \\ &= P\left(\underset{\text{fixed}}{\theta} \in \left[\underset{\text{fixed}}{\hat{\theta}} - \underset{\text{fixed}}{c}, \underset{\text{fixed}}{\hat{\theta}} + \underset{\text{fixed}}{c} \right] \right) \end{aligned}$$

Standard normal quantile

- **Definition.** z_τ is the fixed number such that, for $Z \sim N(0, 1)$:

$$P(Z \leq z_\tau) = \tau, \quad \tau \in (0, 1)$$

- The τ -th quantile of $N(0, 1)$:

$$z_\tau = \Phi^{-1}(\tau), \quad \Phi(z) = P(Z \leq z)$$

- For example:

$$z_{0.975} \approx 1.96, \quad z_{0.025} \approx -1.96$$

- Symmetry around zero:

$$z_\tau = -z_{1-\tau}$$

Choice of c

- Subtract θ and divide by ω :

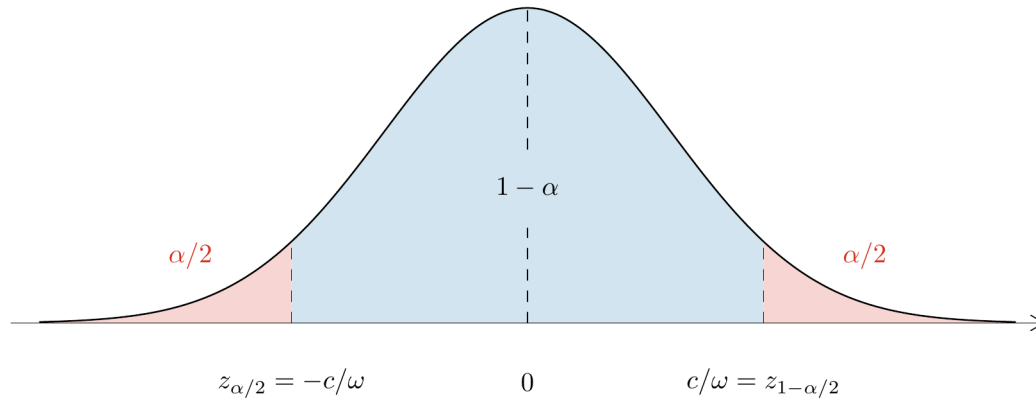
$$\begin{aligned} &\hat{\theta} \sim N(\theta, \omega^2) \\ \Rightarrow &\hat{\theta} - \theta \sim N(0, \omega^2) \\ \Rightarrow &\underbrace{\frac{\hat{\theta} - \theta}{\omega}}_{=Z} \sim N(0, 1) \end{aligned}$$

- Rewrite the event $\theta \in CI_{1-\alpha}$ in terms of Z :

$$\begin{aligned} P(\theta \in CI_{1-\alpha}) &= P(\hat{\theta} - c \leq \theta \leq \hat{\theta} + c) \\ &= P(-c \leq \theta - \hat{\theta} \leq c) \\ &= P(-c \leq \hat{\theta} - \theta \leq c) \\ &= P\left(-\frac{c}{\omega} \leq \frac{\hat{\theta} - \theta}{\omega} \leq \frac{c}{\omega}\right) \\ &= P\left(-\frac{c}{\omega} \leq Z \leq \frac{c}{\omega}\right) \end{aligned}$$

Value of c

- Standard normal density, area $1 - \alpha$ between $-c/\omega$ and c/ω :



- Read c off the picture:

$$\begin{aligned}
 -\frac{c}{\omega} &= z_{\alpha/2} \\
 &= -z_{1-\alpha/2} \\
 \frac{c}{\omega} &= z_{1-\alpha/2} \\
 \Rightarrow c &= z_{1-\alpha/2} \omega
 \end{aligned}$$

Coverage check

- Substitute $c = z_{1-\alpha/2} \omega$:

$$\begin{aligned}
 P(\theta \in CI_{1-\alpha}) &= P(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2}) \\
 &= 1 - \frac{\alpha}{2} - \frac{\alpha}{2} \\
 &= 1 - \alpha
 \end{aligned}$$

where each of the two tails has probability $\alpha/2$.

Interval for θ

- Symmetric interval with coverage $1 - \alpha$:

$$CI_{1-\alpha} = [\hat{\theta} - z_{1-\alpha/2} \omega, \hat{\theta} + z_{1-\alpha/2} \omega]$$

- Centre $\hat{\theta}$, half-length $\omega z_{1-\alpha/2}$.
- For $\alpha = 0.05$:

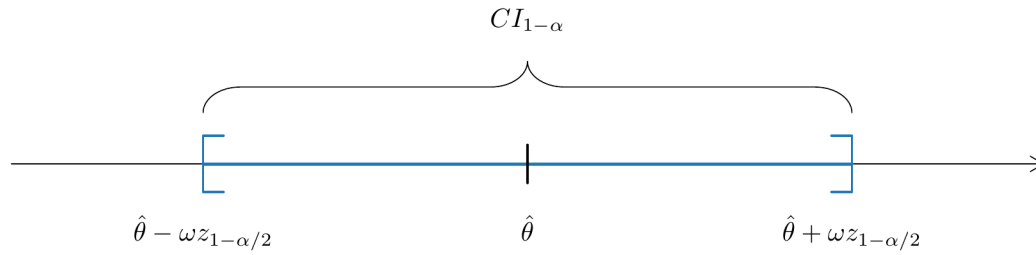
$$z_{1-\alpha/2} = z_{0.975} \approx 1.96$$

- Coverage, written with an absolute value:

$$P(\theta \in CI_{1-\alpha}) = P\left(\left|\frac{\hat{\theta} - \theta}{\omega}\right| \leq z_{1-\alpha/2}\right)$$

Length of $CI_{1-\alpha}$

- $CI_{1-\alpha}$ around $\hat{\theta}$:



- Length:

$$2 \omega z_{1-\alpha/2}$$

- Effect of α :

$$\alpha \downarrow \implies z_{1-\alpha/2} \uparrow$$

Numerical example

- Given:

$$\hat{\theta} = 0.06$$

$$\alpha = 0.05 \implies z_{1-\alpha/2} \approx 2.0$$

$$\omega = 0.01$$

- Substitute into the interval:

$$CI_{1-\alpha} = [\hat{\theta} - z_{1-\alpha/2} \omega, \hat{\theta} + z_{1-\alpha/2} \omega]$$

$$\begin{aligned} CI_{0.95} &= [0.06 - 2.0 \cdot 0.01, 0.06 + 2.0 \cdot 0.01] \\ &= [0.04, 0.08] \end{aligned}$$

- **Question.** What does $CI_{0.95} = [0.04, 0.08]$ mean?

1. $P(0.04 \leq \theta \leq 0.08) = 0.95$
2. $P(0.04 \leq \hat{\theta} \leq 0.08) = 0.95$
3. $P(\theta \in [0.04, 0.08] \mid \text{data}) = 0.95$
4. None of the above

Random and computed interval

- Before $CI_{1-\alpha}$ is computed:

$$P(\theta \in \underbrace{CI_{1-\alpha}}_{\text{random}}) = 1 - \alpha$$

- Once computed:

$$CI_{0.95} = \underbrace{[0.04, 0.08]}_{\text{not random}}$$

Interval for β_1

- With $\hat{\theta} = \hat{\beta}_1$ and $\omega^2 = \sigma^2 / (X_1^\top M_2 X_1)$:

$$CI_{1-\alpha} = \hat{\beta}_1 \pm z_{1-\alpha/2} \sqrt{\frac{\sigma^2}{X_1^\top M_2 X_1}}$$

Other intervals

Asymmetric interval

- Tails of $Z = (\hat{\theta} - \theta) / \omega$: $\alpha/3$ below, $2\alpha/3$ above.
- Interval:

$$\widetilde{CI} = [\hat{\theta} - \omega z_{1-2\alpha/3}, \hat{\theta} + \omega z_{1-\alpha/3}]$$

- Ordering of the quantiles:

$$z_{1-2\alpha/3} < z_{1-\alpha/2} < z_{1-\alpha/3}$$

- For $\alpha = 0.1$:

$$\begin{aligned}\alpha/2 &= 0.05 \\ 1 - \alpha/2 &= 0.95 \\ 1 - \alpha/3 &\approx 0.967 \\ 1 - 2\alpha/3 &\approx 0.933\end{aligned}$$

Coverage of $\widetilde{CI}$

- Coverage of $\widetilde{CI}$ is $1 - \alpha$:

$$\begin{aligned}1 - \alpha &= 1 - \frac{2\alpha}{3} - \frac{\alpha}{3} \\ &= \left(1 - \frac{2\alpha}{3}\right) - P(Z < z_{\alpha/3}) \\ &= P(Z \leq z_{1-2\alpha/3}) - P(Z < z_{\alpha/3}) \\ &= P(z_{\alpha/3} \leq Z \leq z_{1-2\alpha/3}) \\ &= P\left(z_{\alpha/3} \leq \underbrace{\frac{\hat{\theta} - \theta}{\omega}}_{=Z} \leq z_{1-2\alpha/3}\right) \\ &= P(\omega z_{\alpha/3} \leq \hat{\theta} - \theta \leq \omega z_{1-2\alpha/3}) \\ &= P(\omega z_{\alpha/3} - \hat{\theta} \leq -\theta \leq \omega z_{1-2\alpha/3} - \hat{\theta}) \\ &= P(\hat{\theta} - \omega z_{1-2\alpha/3} \leq \theta \leq \hat{\theta} - \omega z_{\alpha/3}) \\ &= P(\hat{\theta} - \omega z_{1-2\alpha/3} \leq \theta \leq \hat{\theta} + \omega z_{1-\alpha/3}) \\ &= P(\theta \in [\hat{\theta} - \omega z_{1-2\alpha/3}, \hat{\theta} + \omega z_{1-\alpha/3}]) \\ &= P(\theta \in \widetilde{CI})\end{aligned}$$

where $z_{\alpha/3} < 0$ and $-z_{\alpha/3} = z_{1-\alpha/3}$.

Length of $\widetilde{CI}$

- Asymmetric and symmetric intervals:

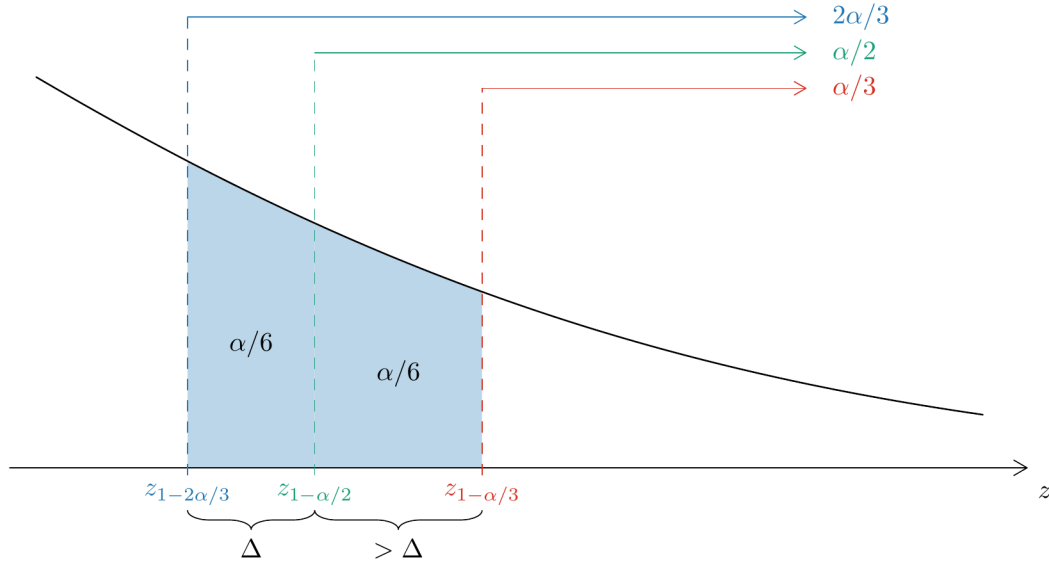
$$\begin{aligned}\text{length of } \widetilde{CI} &= \omega(z_{1-2\alpha/3} + z_{1-\alpha/3}) \\ \text{length of } CI_{1-\alpha} &= \omega(z_{1-\alpha/2} + z_{1-\alpha/2}) \\ &= 2\omega z_{1-\alpha/2}\end{aligned}$$

- The symmetric interval is shorter:

$$2\omega z_{1-\alpha/2} < \omega(z_{1-2\alpha/3} + z_{1-\alpha/3})$$

Upper quantiles

- Right tail of $N(0, 1)$ at $\alpha = 0.1$, with three quantiles and two bands:



- Each band has the same area:

$$\begin{aligned}\frac{2\alpha}{3} - \frac{\alpha}{2} &= \frac{\alpha}{6} \\ \frac{\alpha}{2} - \frac{\alpha}{3} &= \frac{\alpha}{6}\end{aligned}$$

- Density of $N(0, 1)$:

$$\text{pdf}_{N(0,1)}(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

- Gaps between the quantiles:

$$\begin{aligned}z_{1-\alpha/2} - z_{1-2\alpha/3} &= \Delta \\ z_{1-\alpha/3} - z_{1-\alpha/2} &> \Delta\end{aligned}$$

One-sided interval

- Only the lower boundary matters:

$$[t, +\infty)$$

- Interval:

$$[\hat{\theta} - z_{1-\alpha} \omega, +\infty)$$

- Coverage:

$$\begin{aligned} P(\theta \in [\hat{\theta} - z_{1-\alpha} \omega, +\infty)) &= P(\theta \geq \hat{\theta} - z_{1-\alpha} \omega) \\ &= P\left(z_{1-\alpha} \geq \underbrace{\frac{\hat{\theta} - \theta}{\omega}}_{=Z}\right) \\ &= P(Z \leq z_{1-\alpha}) \\ &= 1 - \alpha \end{aligned}$$

Summary

Estimator and standardization

- $\hat{\beta} \mid X \sim N(\beta, \sigma^2(X^\top X)^{-1})$ under Assumptions 1 to 5.
- $\hat{\beta}_1 \mid X \sim N(\beta_1, \sigma^2/(X_1^\top M_2 X_1))$, with $\theta = \beta_1$, $\hat{\theta} = \hat{\beta}_1$, $\omega^2 = \sigma^2/(X_1^\top M_2 X_1)$.
- $\hat{\theta} \sim N(\theta, \omega^2)$ with ω^2 known, and $P(\hat{\theta} = \theta) = 0$.
- $Z = (\hat{\theta} - \theta)/\omega \sim N(0, 1)$; $P(Z \leq z_\tau) = \tau$ and $z_\tau = -z_{1-\tau}$.

Interval and coverage

- $P(\theta \in CI_{1-\alpha}) = 1 - \alpha$ with $\alpha \in (0, 1)$; $CI_{1-\alpha}$ random, its computed value not random.
- $CI_{1-\alpha} = [\hat{\theta} - z_{1-\alpha/2} \omega, \hat{\theta} + z_{1-\alpha/2} \omega]$, of length $2\omega z_{1-\alpha/2}$.
- $\alpha = 0.05$: $z_{0.975} \approx 1.96$; $\alpha \downarrow \implies z_{1-\alpha/2} \uparrow$.
- Regression example: $CI_{1-\alpha} = \hat{\beta}_1 \pm z_{1-\alpha/2} \sqrt{\sigma^2/(X_1^\top M_2 X_1)}$, with $P(\beta_1 \in CI_{1-\alpha} \mid X) = 1 - \alpha$.

Asymmetric and one-sided intervals

- $\widetilde{CI} = [\hat{\theta} - \omega z_{1-2\alpha/3}, \hat{\theta} + \omega z_{1-\alpha/3}]$: coverage $1 - \alpha$.
- Length of $\widetilde{CI}$ against $CI_{1-\alpha}$: $\omega(z_{1-2\alpha/3} + z_{1-\alpha/3}) > 2\omega z_{1-\alpha/2}$.
- $[\hat{\theta} - z_{1-\alpha} \omega, +\infty)$: coverage $1 - \alpha$.